



BTS(C) -V - 05 - 064(G)

B.Tech. Degree V Semester Examination, November 2005

ME 505 HEAT MASS TRANSFER

(2002 Admissions)

Time: 3 Hours

Maximum Marks: 100

- I a) Derive an expression for the heat transfer rate per unit outer surface area of a spherical surface of radii r_1 and r_2 . What is the mean conducting area in this case. (10)
- b) Calculate the heat loss per square metre of outside surface area of a heated sphere 20 cm in diameter covered with 5 cm of insulation ($k=0.07 \text{ W/m K}$). The inside and outside surface temperature of the insulation are 300°C and 50°C respectively. (10)
- OR**
- II a) Explain the term shape factor with reference to heat conduction. (8)
- b) The conductivity of a material varies with temperature as $K = K_0(A + BT + CT^2)$. Find the expression for the mean conductivity of the material. (12)
- III a) Differentiate critically between mechanism of heat transfer by free and forced convection. (10)
- b) 700 Kg of water is heated/min. as it flows through a $25 \times 15 \text{ cm}$ conduit and 8m long. If the water enters at 25°C and leaves the conduit at 35°C , find out the temperature of the conduit surface required. (10)
- OR**
- IV a) List various parameter affecting forced convection heat transfer and their dimensions. Show that these can be grouped into non-dimensional groups. (10)
- b) Explain briefly the physical mechanisms of condensation and boiling. (10)
- V a) Explain the term:
- (i) Emissivity (ii) Absorptivity
- (iii) Transmissivity (2x3=6)
- b) Two parallel rectangular surfaces, $1\text{m} \times 2\text{m}$, are opposite to each other at a distance of 4m. The surfaces are black and at temperatures of 100°C and 200°C respectively. Calculate the rate of heat exchange by radiation between the two surfaces. (14)
- OR**
- VI a) State and prove Stefan-Boltzman law. (6)
- b) Inner walls of a very large furnace are maintained at 1600°C . Outer surface of the walls, which are very thick, is well insulated. A hole 60mm dia, left uncovered, serves on a view point. If the ambient is at 40°C , estimate the energy loss from the port by radiation. (14)

(Turn Over)

- VII a) Derive the expression for the logarithmic mean temperature difference of a parallel flow heat exchanger. (8)
- b) A parallel flow heat exchanger and a counter flow heat exchanger both handle the same weight of same hot fluid (W_H). Both heat exchangers handle equal quantity of water (W_C) which is used on the cold fluid. This gives rise to
- Δt_h , parallel = Δt_h , counter and
 Δt_c , parallel = Δt_c , counter.
- This being the case, what advantages does one obtain by using a counter current heat exchanger. Explain with the help of an example. (12)
- OR**
- VIII a) Explain the meaning of the terms:
- (i) Effectiveness Coefficient
(ii) Heat capacity rates
(iii) Number of heat transfer units (2 x 3 = 6)
- b) In a single pass counter flow heat exchanger 5000 Kg/hr of water enter at 15°C and cools 10,000 kg/hr of an oil having a specific heat of 2.1 KJ/Kg K from 100°C to 65°C. If the overall heat transfer coefficient is 290 W/m²K, determine the surface area required. (14)
- IX a) Define:
- (i) Mass flux
(ii) Mass transfer coefficient
(iii) Diffusion coefficient
(iv) Sherwood number (4 x 2 = 8)
- b) State and explain Fick's Law of Diffusion. (6)
- c) Discuss the analogy between the heat transfer and mass transfer phenomena. (6)
- OR**
- X a) Estimate the diffusion rate of water from the bottom of a test tube 1.5cm in diameter and 15 cm long into dry atmospheric air at 25°C. Assume $D = 0.256 \text{ cm}^2/\text{sec}$. (10)
- b) Hydrogen gas at 2 atm. And 25°C is flowing through a rubber pipe of inner dia = 25 mm and outer dia = 50 mm. The diffusivity of H₂ through rubber is $0.7 \times 10^{-4} \text{ m}^2/\text{hr}$. The solubility of H₂ is $0.053 \text{ cm}^3/\text{cm}^3$ of rubber at 1 atm. Find out the loss of H₂ by diffusion per metre length of the pipe. (10)